



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Post Lab

Aim: Develop a program to implement **First Come First Serve (FCFS) CPU Scheduling Algorithm** for multiple processes. The program takes input for burst time of each process, calculates the **waiting time** and **turnaround time**, and determines the order of execution based on the arrival sequence of processes.

Name:

USN:

Semester / Year:

4th SEM / 2nd YEAR

Academic Session:

2025-26

Date of Performance:

Date of Submission:

❖ **CODE :**

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int n, i;
    float total_wt = 0, total_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    int bt[n], wt[n], tat[n];

    // Input
    printf("Enter Burst Time:\n");
    for(i = 0; i < n; i++) {
        printf("P%d: ", i+1);
        scanf("%d", &bt[i]);
    }

    // Waiting Time
    wt[0] = 0;
    for(i = 1; i < n; i++) {
        wt[i] = wt[i-1] + bt[i-1];
    }

    // Turnaround Time
    for(i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        total_wt += wt[i];
        total_tat += tat[i];
    }

    // Output
    printf("\nProcess\tBT\tWT\tTAT\n");
    for(i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n", i+1, bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f", total_wt/n);
    printf("\nAverage Turnaround Time = %.2f\n", total_tat/n);

    return 0;
}
```

❖ OUTPUT :

```
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ gcc FCFS.c -o FCFS  
  
Tejaswini@LAPTOP-COS64E96 MSYS ~  
$ ./FCFS  
Enter number of processes: 3  
Enter Burst Time:  
P1: 5  
P2: 3  
P3: 8  
  
Process BT      WT      TAT  
P1      5      0      5  
P2      3      5      8  
P3      8      8      16  
  
Average Waiting Time = 4.33  
Average Turnaround Time = 9.67  
Tejaswini@LAPTOP-COS64E96 MSYS ~
```